

GE 501: Creative Engineering Design

Course Code	GE 501
Course Name	Creative Engineering Design
L-T-P-C	2-0-2-3
Intended for	Open to All
Approval	54th BoA (May 2024)

Course Contents

- **Understanding Design:** Introduction to Design Principles and History: Evolution of design, key milestones, and influential figures, Comparison of design across cultures and eras, Terminologies and Definitions Product Life-cycle and Systematic Design Process: Stages of the product lifecycle, Introduction to systematic design processes and methodologies. (6 Hours)
- **Thinking Methods:** Introduction to Design Thinking: Core principles and stages of Design Thinking Empathize: User journey mapping, persona development, empathy exercises, affinity diagram, mind map, Define, Ideate, Prototype, Test. (8 Hours)
- **Engineering Design:** Characteristics of Successful Products: Analysis of market-leading products and their design attributes, Good Design vs Bad Design. (2 Hours)
- **Problem Design:** Opportunity Identification and Customer Needs: Identifying target groups and stakeholders, Techniques for identifying market gaps and acquiring customer feedback. Task Clarification and User Research: Conducting effective user research and habitat studies, Translating customer feedback into user needs Problem Identification and Analysis: Establish Cause-Effect Relationships in user needs, Benchmark existing products - product and patent study, Creating a requirements list and assigning importance to it and SNPS. (8 Hours)
- **Conceptual Design:** Introduction to Conceptual Design: Function structures and the Sapphire Framework Imagination, Motivation, and Inspiration: Techniques for fostering creativity, such as mood boards and bio-inspiration. Ideation Techniques: Brainstorming, Synectics, Random words and other ideation methods Concept Generation and Visualization Tools: Concept generation by combining ideas. Concept Evaluation and Selection: Analytical, simulation, and physical evaluation techniques. (8 Hours)
- **Systematic Embodiment Design:** System Thinking and System Design, Embodiment Design, Detailed Design, Prototyping, Design for Manufacture, Assembly, and Environment (DFMAE) Prototyping Technologies: Introduction to 3D printing, clay, and foam modeling Techniques for rapid prototyping and testing Sketching and Sculpting: Traditional and digital sketching techniques using VR, Basics of sculpting for concept modeling. (10 Hours)

Text Books

1. Chakrabarti, A., **Engineering Design Synthesis**, Springer, 2013.
2. Kaushik Kumar, Muralidhar Kurni, **Design Thinking a Forefront Insight**, CRC Press, 2023.

References

1. Pahl, G., Beitz, W., **Engineering Design: A Systematic Approach, Vol. 3**, Springer, 2007.
2. Dieter, George E.; Schmidt, Linda C., **Engineering Design**, 5th Edition, McGraw-Hill Education, 2012.
3. Pressman, Andrew, **Design Thinking: A Guide to Creative Problem Solving for Everyone**, Routledge/Productivity Press, 2019.
4. Lewrick, Michael, **Design Thinking and Innovation Metrics**, Wiley Publications, 2023 .
5. Brown, Tim, **Change by Design**, Harper Collins Publisher, 2020.
6. Eriksson, Yvonne, **Different Perspectives in Design Thinking**, CRC Press, 2022.
7. Muller-Rotterberg, Christian, **Design Thinking for Dummies**, John Wiley & Sons, 2020.
8. Cross, N., **Engineering Design Methods**, Vol. 4, John Wiley and Sons Ltd., 2008.
9. Reid, K., Estell, J., **Engineering Design and the Product Lifecycle**, Momentum Press Engineering, 2018.